

Data Management Technologies

Unit 7 - DMT - individual consolidation task

Activity 1. Data Model Identification

The data sets analysed are described below:

The datasets relate to data for a fictional school, called Greenfield Academy.

File	Type	Description / Classification justification
students.csv	Relational table	Contains student information (ID, name, DOB and year group). The information is presented in a table containing rows and columns, and as such is a relational model.
classes.csv	Relational table	Contains lesson information (teacher, room, day, time). The information is presented in a table containing rows and columns, and as such is a relational model.
enrollments.csv	Relational table	This table links information from the above table (students and classes), and gives information on which students are enrolled in which classes.
timetable.xml	XML	This document is a school timetable in a hierarchical structure. Each class is contained in a time-slot, which is contained within a day-slot.
noticeboard.json	Document	The noticeboard contains 5 posts with notes/announcements. It follows the document model, with each post being self-contained and semi-structured.

Activity 2. XML and JSON Syntax

Given a small textual dataset (e.g., a list of books with titles, authors, and publication years), students must:

1. Create an XML representation.
2. Create a JSON representation.

The output of the subtasks above will be two separate files containing the XML and JSON representations saved in your OneDrive, with or without a link provided.

Links:

XML representation link: [books.xml](#)

JSON representation link: [books.json](#)

The XML and JSON files are also in the corresponding GitHub folder (Unit 7 Individual Consolidation Task Data – Activity 2)

Activity 3. Exploring TinyDB with Python

Install TinyDB and set up a small database in JSON format using Python.

Perform the following queries:

1. Insert a new record.
2. Retrieve all records where a specific attribute matches a value.
3. Update an existing record.

Your output will be Python script files with the # Queries and a short explanation of each query by way of comment

(Completed in separate Python script file)